

Richard Teague

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ACADEMIC APPOINTMENTS

07/26–present	Kerr-McGee Development Associate Professor MIT, Department of Earth, Atmospheric and Planetary Sciences
07/22–06/26	Kerr-McGee Development Assistant Professor MIT, Department of Earth, Atmospheric and Planetary Sciences
05/22–04/25	Research Associate Smithsonian Astrophysical Observatory
09/19–04/22	Submillimeter Array Fellow Center for Astrophysics · Harvard & Smithsonian
05/17–07/19	Postdoctoral Researcher University of Michigan
01/17–04/17	Postdoctoral Researcher Max-Planck-Institute for Astronomy

EDUCATION

2013–2017	Ph.D. in Astronomy (Magna Cum Laude) Max-Planck-Institute for Astronomy, Heidelberg, Germany
2008–2013	MPhys Astrophysics (First Class Honours) University of Edinburgh, Edinburgh, United Kingdom

AWARDS & HONORS

2026	NSF CAREER Revealing the Influence of Magnetic Fields During the Formation of Planetary Systems
2026	AAS Fred Kavli Plenary Speaker For pioneering work revealing the 3D kinematic velocity fields within planet-forming disks
2024	EAPS Community Builder Award Recognition of an EAPS community member for exceptional service to EAPS

- 2022** **pH Lectureship**
Recognizes a CfA scientist who shows exceptional promise early in their career
- 2016** **Ernst Patzer Award**
Awarded for the best refereed publication by a young scientist
- 2010, 2011** **Pre-Honours Certificate of Merit**
Top 5% performance in pre-honours exams, University of Edinburgh

OBSERVATIONAL TIME & TELESCOPE PROPOSALS (AS PI)

Over 392 hours (600+ as co-I) awarded on ALMA as PI, including the exoALMA Large Program; 20 hours (165 as co-I) on IRAM; 46 hours (30 as co-I) on the SMA; 26 hours (18 as co-I) on JWST as co-PI; 6 hours on VLT/GRAVITY; 3 nights (9 as co-I) on Magellan. Selected PI proposals below.

- 2025** **Detecting and Characterizing Magnetic Fields in Two Disks through Zeeman Splitting**
ALMA, 47 hours, 2025.1.00304.S, B ranked · PI
- 2025** **Calibrating Disk Mass Measurement Techniques**
ALMA, 14 hours, 2025.1.00940.S, A ranked · PI
- 2022** **The Most Sensitive Search for Magnetic Fields in a Solar Nebula Analogue**
ALMA, 18 hours, 2022.1.00840.S, A ranked · PI
- 2021** **exoALMA Large Program**
ALMA, 183 hours, 2021.1.01123.L, A ranked · PI (co-PIs: Benisty, Facchini, Fukagawa & Pinte)
- 2021** **Is the Magneto-Rotational Instability Driving Protoplanetary Disk Evolution?**
SMA, 30 hours, 2020A-S033, A ranked · PI
- 2019** **Mapping the 3D Kinematic Structure of Planet Formation**
ALMA, 33.2 hours, 2019.1.00419.S, B ranked · PI
- 2018** **An Unambiguous Detection of a Magnetic Field in a Protoplanetary Disk**
ALMA, 6.7 hours, 2018.1.00980.S, A ranked · PI

INVITED TALKS & COLLOQUIA

Talks and colloquia since 2015, reverse chronological

- 02/27** **“What Drives Planet Formation?”**
ALMA at 15 Years - Science, Synergies and the Road Ahead
- 01/27** **“Reconstructing the Planet Formation Environment”**
Biomedical and Astronomical Signal Processing Frontiers

- 06/26 **“Dynamics of Planet Formation”**
Perspectictives of Planet and Star Formation
- 04/26 **“Open Challenges in the Modeling and Interpretation of Protoplanetary Disk Observations”**
ML4Disks Workshop
- 03/26 **“From Pebbles to Planets: New Frontiers in Planet Formation”**
Caltech GPS Division Seminar
- 02/26 **“Searching for Turbulence: Methods, Limitations and Potential”**
Flatiron Workshop on Hydrodynamic and Dusty Turbulence in Protoplanetary Disks
- 01/26 **“The Search for Magnetic Fields in Protoplanetary Disks”**
MPIA Star and Planet Formation Seminar
- 01/26 **“The Search for Magnetic Fields in Protoplanetary Disks”**
ESO Star and Planet Formation Seminar
- 05/25 **“An Overview of the exoALMA Program”**
COST Action: PLANETS Seminar
- 04/25 **“How To Build a Solar System”**
Johns Hopkins & STScI Joint Colloquium
- 03/25 **“How To Build a Solar System”**
UC Berkeley Astronomy Colloquium
- 06/24 **“An Interferometric View of the Planet Formation Environment: Past, Present and Future”**
GRAVITY+: Impact on Star and Planet Formation
- 03/24 **“From Pebbles to Planets: What Gas Can Tell Us”**
McMaster Astronomy Colloquium
- 03/24 **“From Pebbles to Planets: What Gas Can Tell Us”**
Canadian Institute for Theoretical Astrophysics Colloquium
- 03/24 **“Molecular Probes of the Planet Formation Environment”**
Star to Planet Formation Workshop
- 02/24 **“From Pebbles to Planets: What Gas Can Tell Us”**
Munich Joint Astronomical Colloquium
- 02/24 **“exoALMA: Early Results and Future Outlook”**
Munich University Astronomy Colloquium
- 12/23 **“The Dynamical Structure of Planet Forming Disks”**
ALMA at 10 Years: Past, Present, and Future
- 06/23 **“Witnessing the Formation of Giant Planets and their Moons”**
Gordon Conference on the Origins of Solar Systems

- 05/23 **“Witnessing the Earliest Stages of Planet Formation”**
MATH + X: Planet Formation and Habitability
- 05/23 **“Witnessing the Formation of Giant Planets and their Moons”**
Boston University Astrophysics Seminar
- 04/23 **“Witnessing the Formation of Giant Planets and their Moons”**
MIT Haystack Colloquium
- 03/23 **“Witnessing the Formation of Giant Planets and their Moons”**
Ohio State University Astronomy Colloquium
- 02/23 **“Witnessing the Formation of Giant Planets and their Moons”**
Harvard University Department of Earth and Planetary Sciences Colloquium
- 10/22 **“ALMA’s 3D View of Planet Formation”**
From Clouds to Planets II: The Astrochemical Link
- 09/22 **“Exploring the Youngest Planetary Systems”**
Center for Astrophysics | Harvard & Smithsonian pH Lecture
- 02/22 **“Detecting the Youngest Planets”**
University of Florida Astronomy Colloquium
- 02/22 **“Detecting the Youngest Planets”**
Penn State CEHW Seminar Series
- 11/21 **“Detecting Molecular Line Polarization in Protoplanetary Disks”**
Pan-Experiment Galactic Science Group Seminar Series
- 10/21 **“Mapping the Assembly of Planetary Systems in 6 Dimensions”**
Munich Joint Astronomical Colloquium
- 09/21 **“Mapping the Assembly of Planetary Systems in 6 Dimensions”**
Center for Astrophysics | Harvard & Smithsonian Colloquium
- 05/21 **“Witnessing the Assembly of Planetary Systems”**
ETH Zurich Exoplanets & Habitability Seminar
- 05/21 **“Witnessing the Assembly of Planetary Systems”**
Cambridge Exoplanet Center Seminar
- 04/21 **“Transforming ALMA into a Planet Hunting Facility”**
Towards the Comprehensive Characterization of Exoplanets: Science at the Interface of Multiple Measurement Techniques
- 04/21 **“Witnessing the Assembly of Planetary Systems”**
McMaster University Astrophysics Seminar
- 03/21 **“Observations and Observational Predictions”**
Circumplanetary Disks II

- 02/21 **“Witnessing the Assembly of Planetary Systems”**
Max Planck Research Group Selection Symposium
- 02/21 **“Planet Formation in Six Dimensions”**
Caltech Dix Planetary Science Department Seminar
- 12/20 **“Observing the Kinematics of Gaseous Substructures”**
Five Years After HL Tau: A New Era in Planet Formation
- 10/20 **“Observing the Dynamics of Planet Disk Interactions”**
Research Unit Transition Disks (RUTD) Conference
- 07/20 **“Kinematical Detection and Characterizing of Protoplanets with ALMA”**
Exoplanets III
- 07/20 **“Visualizing the Assembly of Planetary Systems”**
MPIA Königstuhl Colloquium
- 11/19 **“Witnessing the Dynamics of Planetary Assembly”**
JPL Astrophysics Colloquium
- 10/19 **“Exploiting ALMA’s Potential for Planet Hunting”**
Visualizing the Kinematics of Planet Formation
- 06/19 **“Unveiling the Dynamics of Planet Formation”**
Gordon Research Seminar
- 04/19 **“The Physical Conditions of Planet Formation with Molecular Excitation”**
IAU Symposium 350: Laboratory Astrophysics
- 03/19 **“Unveiling the Dynamics of Planet Formation”**
Planet-Forming Disks
- 10/18 **“Observing the Kinematics of Planet-Disk Interactions with ALMA”**
NAOJ Theoretical Astronomy Seminar
- 08/18 **“Using Kinematics to Search for Embedded Protoplanets”**
LMU Munich Astronomy Colloquium
- 08/18 **“Kinematical Detections of Embedded Protoplanets”**
University of Tübingen Astronomy Seminar
- 04/18 **“The First Kinematic Detection of Embedded Protoplanets”**
Astrophysical Frontiers in the Next Decade and Beyond
- 02/18 **“A Spatially Resolved Search for Turbulence in TW Hya”**
Magnetic Fields or Turbulence
- 11/16 **“Measuring Turbulence in TW Hya with ALMA: Methods and Limitations”**
MPIA Patzer Awards Colloquium

11/16	“Observing the Earliest Stages of Planet Formation” MPIA Königstuhl Colloquium
06/16	“Detecting Turbulence in Protoplanetary Disks” Astrochemistry with ALMA Cycle 4
04/16	“Turbulence in Protoplanetary Disks: Methods and Limitations” Sant-Cugat Forum on Astrophysics
03/16	“Turbulence in TW Hya” Protoplanetary Discussions
01/15	“Deuterium Fraction in Protoplanetary Disks” Chemical Diagnostics of Star and Planet Formation
01/15	“Deuterium Fraction in DM Tau” ZAG – IPAG – MPIA Workshop on Planet Formation

TEACHING

2022–present	12.410 / 8.287: Observational Techniques for Optical Astronomy Instructor
2025–present	12.423 / 12.623: Planet Formation Instructor
2024	12.S681: From Grains to Gas Giants: The Formation of Planetary Systems Instructor
2014	Wavefront Analysis Laboratory Instructor

ADVISING & MENTORING

All students and postdocs are based at MIT unless otherwise noted

Ph.D. students	Leah Albrow (2024–present) · Isabella Macias (2024–present) · Jensen Pierce Thomas Lawrence (2023–present)
MSc. students	Marco Duraku, Imperial College London (2025–2026, MIT–Imperial Exchange) · Amius Marshall De’Ath (2024–2025, MIT–Imperial Exchange) · Abigail Colclasure (2024–2025)
Undergraduates	Headam Im (2026–present) · Jackson Holland (2026) · Erin Cusson (2024–2026) · Anika Nath (2024) · Carol Chen (2022–2023) · Aidan van Duzer (2023) · Anna Orgel (2022–2023)

Academic advisor	Lily Jones (2026–present) · Jan Toomlaid (2024–2026) · Kaylee Barrera (2023–2026) · Anika Nath (2023–2025)
Prior institutions	Co-supervised graduate and undergraduate students at the University of Michigan (Felipe Alcaron, Jenny Calahan, Deryl Long, Case Hazewinkel, Jeanne Kwon, 2018–2020), Harvard University (Alessandra Canta, 2020–2021), Beijing Normal University (Haochuan Yu, 2020–2022) and Ludwig Maximilian University (Julian Penzinger, 2016, 2018)

PROFESSIONAL SERVICE

Refereeing	Referee for <i>AAS</i> , <i>A&A</i> , <i>MNRAS</i> and <i>Nature</i> journals · NESSF External Reviewer · European Research Council External Reviewer
Departmental (MIT/EAPS)	EAPS Council, Junior Faculty Representative (2025–present) · CORE: Community, Outreach and Respect in EAPS (2025–present) · EAPS Department Lecture Series Planetary Science Representative (2025–2026) · EAPS Committee on the Education Program (2023–2024) · EAPS Distinguished Postdoctoral Fellowship Committee (2022, 2025)
Workshop & conference SOCs	Astrochemistry in the Broadband Era: ngVLA and ALMA WSU (2025) · Spatio-spectral Modeling of ALMA Data Cubes: ALMA-2030 (2024) · exoALMA Start of Science Workshop (2022) · Vertical Shear Instability Meeting (2022) · SMA Interferometry School (2021) · Visualizing the Kinematics of Planet Formation (2019)
Diversity, equity & inclusion	Postdoc and Research Scientist DEI Representative (2018–2019) · Conversations on Equity and Inclusion Co-organizer (2018–2019) · Equi-Tea Organizer (2018–2019)
Seminar organizing	SMA Seminar Organizer (2020–2021) · Stars, Planets and Formation Seminar Organizer, UMich (2018–2019) · MPIA Student Representative & Workshop Organizer (2015–2017) · IMPRS Graduate Student Representative (2013–2017)
Outreach	“How to Find Baby Planets,” University of Michigan Lowbrow Astronomers (2020)

REFEREED PUBLICATIONS

179 refereed articles · 24 as first author · reverse chronological · titles are links to paper

2026

20 PAPERS

- 179 **First Detection of HC₅N in a Class II Disk around TW Hya**
Wampler, S. C., R. A. Loomis, A. Diop, et al.
The Astrophysical Journal, 1003, L30

- 178 **An SMA Molecular Inventory of the Edge-on Protoplanetary Disk Gomez's Hamburger**
Cusson, E. M., L. Wölfer, **R. Teague**, et al.
arXiv e-prints, arXiv:2605.26512
- 177 **The circumbinary disc of HD 34700A: II. Analysis of a strong dust asymmetry**
Fasano, D., M. Benisty, J. Stadler, et al.
Astronomy and Astrophysics, 709, A174
- 176 **Probing Dust in the MWC 480 Disk from Millimeter to Centimeter Wavelengths**
Shi, Y., F. Long, E. Macías, et al.
The Astrophysical Journal, 1001, 27
- 175 **exoALMA. XXIV. Formaldehyde Emission in Protoplanetary Disks of exoALMA Compared with Their Properties and Dynamical State**
Alarcón, F., S. Facchini, L. Trapman, et al.
The Astrophysical Journal, 1000, L32
- 174 **exoALMA. XXIII. Estimating Disk and Planet Properties from Dust Morphologies with DBNets 2.0**
Ruzza, A., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 1000, L16
- 173 **exoALMA XXII: A Two-dimensional Atlas of Deviations from Keplerian Disks**
Fukagawa, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L15
- 172 **exoALMA. XXI. The Morphology and Dynamics of Vertical Flows**
Benisty, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L14
- 171 **exoALMA. XX. Tomographic Detection of Embedded Planets in Protoplanetary Disks**
Izquierdo, A. F., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 1000, L13
- 170 **A Protoplanet Candidate in the PDS 66 Disk Indicated by Silicon Sulfide Isotopologues**
Yoshida, T. C., F. Alarcón, J. Bae, et al.
The Astrophysical Journal, 999, L22
- 169 **Benchmarking pre-main sequence stellar evolutionary tracks using disk-based dynamical stellar masses**
Zallio, L., M. Vioque, S. M. Andrews, et al.
Astronomy and Astrophysics, 708, L1

- 168 **AB Aur, a Rosetta stone for studies of planet formation: IV. C/O estimates from CS and SO interferometric observations**
Rivière-Marichalar, P., A. Fuente, R. le Gal, et al.
Astronomy and Astrophysics, 707, A348
- 167 **The circumbinary disk of HD34700A: I. CO gas kinematics indicate spirals, infall, and vortex motions**
Stadler, J., M. Benisty, F. Zagaria, et al.
Astronomy and Astrophysics, 707, A160
- 166 **A 2 au Resolution View by ALMA of the Planet-hosting WISPIT 2 Disk**
Facchini, S., P. Curone, M. Benisty, et al.
The Astrophysical Journal, 998, L16
- 165 **Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement**
Armitage, T., J. Williams, K. Zhang, et al.
The Astrophysical Journal, 998, 308
- 164 **exoALMA. XIX. Confirmation of Non-thermal Line Broadening in the DM Tau Protoplanetary Disk**
Hardiman, C., C. Pinte, D. J. Price, et al.
The Astrophysical Journal, 997, L47
- 163 **Physical and Chemical Characterization of GY 91's Multi-ringed Protostellar Disk with ALMA**
Jiang, S. D., J. Huang, I. Czekala, et al.
arXiv e-prints, arXiv:2601.18884
- 162 **A Submillimeter Survey of CS Excitation in Protoplanetary Disks: Evidence of X-Ray-driven Sulfur Chemistry**
Law, C. J., R. Le Gal, K. I. Öberg, et al.
The Astrophysical Journal, 997, 91
- 161 **Astrometric view of companions in the inner dust cavities of protoplanetary discs**
Vioque, M., R. A. Booth, E. Ragusa, et al.
Astronomy and Astrophysics, 705, A238
- 160 **The ^{12}CO gas structures of protoplanetary disks in the Upper Scorpius region**
Zallio, L., G. P. Rosotti, M. Vioque, et al.
Astronomy and Astrophysics, 705, A49
- 159 **Spirals and Vertical Motions in the Planet-Forming Disk around HD 100546. A multi-line study of its gas kinematics**
Wölfer, L., A. F. Izquierdo, A. Booth, et al.
arXiv e-prints, arXiv:2512.13814

- 158 **Direct Measurement of Extinction in a Planet-hosting Gap**
Cugno, G., S. Facchini, F. Alarcon, et al.
The Astronomical Journal, 170, 317
- 157 **Satellites and Small Bodies With ALMA: Insights Into Solar System Formation and Evolution**
de Kleer, K., M. E. Brown, M. Cordiner, et al.
AGU Advances, 6, e2025AV001778
- 156 **Winding motion of spirals in a gravitationally unstable protoplanetary disk**
Yoshida, T. C., H. Nomura, K. Doi, et al.
Nature Astronomy, 9, 1672
- 155 **The Light Echo of a High-redshift Quasar Mapped with Ly α Tomography**
Eilers, A., M. Yue, J. Matthee, et al.
The Astrophysical Journal, 991, L40
- 154 **JWST-MIRI Observations of the Irradiated Chemistry in the Inner Disk Cavity of GM Aur**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 991, 128
- 153 **Revealing Fine Structure in Protoplanetary Disks with Physics Constrained Neural Fields**
Levis, A., N. Luong, **R. Teague**, et al.
arXiv e-prints, arXiv:2509.03623
- 152 **A Radially Resolved Magnetic Field Threading the Disk of TW Hya**
Teague, R., B. Lankhaar, S. M. Andrews, et al.
The Astrophysical Journal, 991, L6
- 151 **exoALMA. XVIII. Interpreting Large-scale Kinematic Structures as Moderate Warping**
Winter, A. J., M. Benisty, A. F. Izquierdo, et al.
The Astrophysical Journal, 990, L10
- 150 **Leaky dust trap in the PDS 70 disc revealed by ALMA Band 9 observations**
Sierra, A., M. Benisty, P. Pinilla, et al.
Monthly Notices of the Royal Astronomical Society, 541, 3101
- 149 **SO Emission in the Dynamically Perturbed Protoplanetary Disks around CQ Tau and MWC 758**
Zagaria, F., H. Jiang, G. Cataldi, et al.
The Astrophysical Journal, 989, 30
- 148 **Inner disc and circumplanetary material in the PDS 70 system: Insights from multi-epoch, multi-frequency ALMA observations**
Fasano, D., M. Benisty, P. Curone, et al.
Astronomy and Astrophysics, 699, A373

- 147 **Radial variations in the nitrogen, carbon, and hydrogen fractionation in the PDS 70 planet-hosting disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 698, A115
- 146 **exoALMA. XVII. Characterizing the Gas Dynamics around Dust Asymmetries**
Wölfer, L., M. Barraza-Alfaro, **R. Teague**, et al.
The Astrophysical Journal, 984, L22
- 145 **exoALMA. XVI. Predicting Signatures of Large-scale Turbulence in Protoplanetary Disks**
Barraza-Alfaro, M., M. Flock, W. Béthune, et al.
The Astrophysical Journal, 984, L21
- 144 **exoALMA. XV. Interpreting the Height of CO Emission Layer**
Rosotti, G. P., C. Longarini, T. Paneque-Carreño, et al.
The Astrophysical Journal, 984, L20
- 143 **exoALMA. XIV. Gas Surface Densities in the RX J1604.3–2130 A Disk from Pressure-broadened CO Line Wings**
Yoshida, T. C., P. Curone, J. Stadler, et al.
The Astrophysical Journal, 984, L19
- 142 **exoALMA. XIII. Gas Masses from N_2H^+ and C^{18}O : A Comparison of Measurement Techniques for Protoplanetary Gas Disk Masses**
Trapman, L., C. Longarini, G. P. Rosotti, et al.
The Astrophysical Journal, 984, L18
- 141 **exoALMA. XII. Weighing and Sizing exoALMA Disks with Rotation Curve Modelling**
Longarini, C., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 984, L17
- 140 **exoALMA. XI. ALMA Observations and Hydrodynamic Models of LkCa 15: Implications for Planetary Mass Companions in the Dust Continuum Cavity**
Gardner, C. H., A. Isella, H. Li, et al.
The Astrophysical Journal, 984, L16
- 139 **exoALMA. X. Channel Maps Reveal Complex ^{12}CO Abundance Distributions and a Variety of Kinematic Structures with Evidence for Embedded Planets**
Pinte, C., J. D. Ilee, J. Huang, et al.
The Astrophysical Journal, 984, L15
- 138 **exoALMA. IX. Regularized Maximum Likelihood Imaging of Non-Keplerian Features**
Zawadzki, B., I. Czekala, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L14

- 137 **exoALMA. VIII. Probabilistic Moment Maps and Data Products Using Nonparametric Linear Models**
Hilder, T., A. R. Casey, D. J. Price, et al.
The Astrophysical Journal, 984, L13
- 136 **exoALMA. VII. Benchmarking Hydrodynamics and Radiative Transfer Codes**
Bae, J., M. Flock, A. Izquierdo, et al.
The Astrophysical Journal, 984, L12
- 135 **exoALMA. VI. Rotating under Pressure: Rotation Curves, Azimuthal Velocity Substructures, and Gas Pressure Variations**
Stadler, J., M. Benisty, A. J. Winter, et al.
The Astrophysical Journal, 984, L11
- 134 **exoALMA. V. Gaseous Emission Surfaces and Temperature Structures**
Galloway-Sprietsma, M., J. Bae, A. F. Izquierdo, et al.
The Astrophysical Journal, 984, L10
- 133 **exoALMA. IV. Substructures, Asymmetries, and the Faint Outer Disk in Continuum Emission**
Curone, P., S. Facchini, S. M. Andrews, et al.
The Astrophysical Journal, 984, L9
- 132 **exoALMA. III. Line-intensity Modeling and System Property Extraction from Protoplanetary Disks**
Izquierdo, A. F., J. Stadler, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L8
- 131 **exoALMA. II. Data Calibration and Imaging Pipeline**
Loomis, R. A., S. Facchini, M. Benisty, et al.
The Astrophysical Journal, 984, L7
- 130 **exoALMA. I. Science Goals, Project Design, and Data Products**
Teague, R., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 984, L6
- 129 **Mapping the Merging Zone of Late Infall in the AB Aur Planet-forming System**
Speedie, J., R. Dong, **R. Teague**, et al.
The Astrophysical Journal, 981, L30
- 128 **Chemistry in the GG Tau A Disk: Constraints from H_2D^+ , N_2H^+ , and DCO^+ High Angular Resolution ALMA Observations**
Kashyap, P., L. Majumdar, A. Dutrey, et al.
The Astrophysical Journal, 976, 258

- 127 **Kinematical signatures: Distinguishing between warps and radial flows**
Zuleta, A., T. Birnstiel, & **R. Teague**
Astronomy and Astrophysics, 692, A56
- 126 **Observational signatures of circumbinary discs - II. Kinematic signatures in velocity residuals**
Calcino, J., B. J. Norfolk, D. J. Price, et al.
Monthly Notices of the Royal Astronomical Society, 534, 2904
- 125 **Detection of Dimethyl Ether in the Central Region of the MWC 480 Protoplanetary Disk**
Yamato, Y., Y. Aikawa, V. V. Guzmán, et al.
The Astrophysical Journal, 974, 83
- 124 **Gravitational instability in a planet-forming disk**
Speedie, J., R. Dong, C. Hall, et al.
Nature, 633, 58
- 123 **ALMA high-resolution observations unveil planet formation shaping molecular emission in the PDS 70 disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 689, A65
- 122 **Outflow Driven by a Protoplanet Embedded in the TW Hya Disk**
Yoshida, T. C., H. Nomura, C. J. Law, et al.
The Astrophysical Journal, 971, L15
- 121 **On Kinematic Measurements of Self-gravity in Protoplanetary Disks**
Andrews, S. M., **R. Teague**, C. P. Wirth, et al.
The Astrophysical Journal, 970, 153
- 120 **Constraining the stellar masses and origin of the protostellar VLA 1623 system**
Sadavoy, S. I., P. Sheehan, J. J. Tobin, et al.
Astronomy and Astrophysics, 687, A308
- 119 **The First Spatially Resolved Detection of ^{13}CN in a Protoplanetary Disk and Evidence for Complex Carbon Isotope Fractionation**
Yoshida, T. C., H. Nomura, K. Furuya, et al.
The Astrophysical Journal, 966, 63
- 118 **The Carbon Isotopic Ratio and Planet Formation**
Bergin, E. A., A. Bosman, **R. Teague**, et al.
The Astrophysical Journal, 965, 147
- 117 **Mapping the Vertical Gas Structure of the Planet-hosting PDS 70 Disk**
Law, C. J., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 964, 190

2023

17 PAPERS

- 116 **High turbulence in the IM Lup protoplanetary disk. Direct observational constraints from CN and C₂H emission**
Paneque-Carreño, T., A. F. Izquierdo, **R. Teague**, et al.
Astronomy and Astrophysics, 684, A174
- 115 **JWST-MIRI Spectroscopy of Warm Molecular Emission and Variability in the AS 209 Disk**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 964, 36
- 114 **Protoplanetary disks in K_s-band total intensity and polarized light**
Ren, B. B., M. Benisty, C. Ginski, et al.
Astronomy and Astrophysics, 680, A114
- 113 **MAPS: Constraining Serendipitous Time Variability in Protoplanetary Disk Molecular Ion Emission**
Waggoner, A. R., L. I. Cleaves, R. A. Loomis, et al.
The Astrophysical Journal, 956, 103
- 112 **A Magnetically Driven Disk Wind in the Inner Disk of PDS 70**
Campbell-White, J., C. F. Manara, M. Benisty, et al.
The Astrophysical Journal, 956, 25
- 111 **MagAO-X and HST High-contrast Imaging of the AS209 Disk at H α**
Cugno, G., Y. Zhou, T. Thanathibodee, et al.
The Astronomical Journal, 166, 162
- 110 **Three-dimensional magnetic field imaging of protoplanetary disks using Zeeman broadening and linear polarization observations**
Lankhaar, B. & **R. Teague**
Astronomy and Astrophysics, 678, A17
- 109 **Constraining the gas distribution in the PDS 70 disc as a method to assess the effect of planet-disc interactions**
Portilla-Revelo, B., I. Kamp, S. Facchini, et al.
Astronomy and Astrophysics, 677, A76
- 108 **Implications for Chondrule Formation Regions and Solar Nebula Magnetism from Statistical Reanalysis of Chondrule Paleomagnetism**
Fu, R. R., S. C. Steele, J. B. Simon, et al.
The Planetary Science Journal, 4, 151
- 107 **Observational signatures of circumbinary discs - I. Kinematics**
Calcino, J., D. J. Price, C. Pinte, et al.
Monthly Notices of the Royal Astronomical Society, 523, 5763
- 106 **Kinematic Structures in Planet-Forming Disks**
Pinte, C., **R. Teague**, K. Flaherty, et al.
Protostars and Planets VII, 534, 645

- 105 **Tentative co-orbital submillimeter emission within the Lagrangian region L₅ of the protoplanet PDS 70 b**
Balsalobre-Ruza, O., I. de Gregorio-Monsalvo, J. Lillo-Box, et al.
Astronomy and Astrophysics, 675, A172
- 104 **Molecules with ALMA at Planet-forming Scales (MAPS): Complex Kinematics in the AS 209 Disk Induced by a Forming Planet and Disk Winds**
Galloway-Sprietsma, M., J. Bae, **R. Teague**, et al.
The Astrophysical Journal, 950, 147
- 103 **An infrared transient from a star engulfing a planet**
De, K., M. MacLeod, V. Karambelkar, et al.
Nature, 617, 55
- 102 **Mapping Protoplanetary Disk Vertical Structure with CO Isotopologue Line Emission**
Law, C. J., **R. Teague**, K. I. Öberg, et al.
The Astrophysical Journal, 948, 60
- 101 **A kinematically detected planet candidate in a transition disk**
Stadler, J., M. Benisty, A. Izquierdo, et al.
Astronomy and Astrophysics, 670, L1
- 100 **Kinematics and brightness temperatures of transition discs. A survey of gas substructures as seen with ALMA**
Wölfer, L., S. Facchini, N. van der Marel, et al.
Astronomy and Astrophysics, 670, A154
- 99 **UV-driven chemistry as a signpost of late-stage planet formation**
Calahan, J. K., E. A. Bergin, A. D. Bosman, et al.
Nature Astronomy, 7, 49
- 98 **Cold Deuterium Fractionation in the Nearest Planet-forming Disk**
Romero-Mirza, C. E., K. I. Öberg, C. J. Law, et al.
The Astrophysical Journal, 943, 35
- 97 **A kinematic excess in the annular gap and gas-depleted cavity in the disc around HD 169142**
Garg, H., C. Pinte, I. Hammond, et al.
Monthly Notices of the Royal Astronomical Society, 517, 5942
- 96 **A Localized Kinematic Structure Detected in Atomic Carbon Emission Spatially Coincident with a Proposed Protoplanet in the HD 163296 Disk**
Alarcón, F., E. A. Bergin, & **R. Teague**
The Astrophysical Journal, 941, L24

- 95 **Unveiling the outer dust disc of TW Hya with deep ALMA observations**
Ilee, J. D., C. Walsh, J. Jennings, et al.
Monthly Notices of the Royal Astronomical Society, 515, L23
- 94 **Mapping the Complex Kinematic Substructure in the TW Hya Disk**
Teague, R., J. Bae, S. M. Andrews, et al.
The Astrophysical Journal, 936, 163
- 93 **Molecules with ALMA at Planet-forming Scales (MAPS): A Circumplanetary Disk Candidate in Molecular-line Emission in the AS 209 Disk**
Bae, J., **R. Teague**, S. M. Andrews, et al.
The Astrophysical Journal, 934, L20
- 92 **CO Line Emission Surfaces and Vertical Structure in Midinclination Protoplanetary Disks**
Law, C. J., S. Crystian, **R. Teague**, et al.
The Astrophysical Journal, 932, 114
- 91 **Gas Disk Sizes from CO Line Observations: A Test of Angular Momentum Evolution**
Long, F., S. M. Andrews, G. Rosotti, et al.
The Astrophysical Journal, 931, 6
- 90 **Gas and Dust Shadows in the TW Hydrae Disk**
Teague, R., J. Bae, M. Benisty, et al.
The Astrophysical Journal, 930, 144
- 89 **Polarization from Aligned Dust Grains in the β Pic Debris Disk**
Hull, C. L. H., H. Yang, P. C. Cortés, et al.
The Astrophysical Journal, 930, 49
- 88 **Probing inner and outer disk misalignments in transition disks. Constraints from VLTI/GRAVITY and ALMA observations**
Bohn, A. J., M. Benisty, K. Perraut, et al.
Astronomy and Astrophysics, 658, A183
- 87 **Discovery of Molecular-line Polarization in the Disk of TW Hya**
Teague, R., C. L. H. Hull, S. Guilloteau, et al.
The Astrophysical Journal, 922, 139
- 86 **disksurf: Extracting the 3D Structure of Protoplanetary Disks**
Teague, R., C. Law, J. Huang, et al.
The Journal of Open Source Software, 6, 3827
- 85 **Molecules with ALMA at Planet-forming Scales. XX. The Massive Disk around GM Aurigae**
Schwarz, K. R., J. K. Calahan, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 20

2021

40 PAPERS

- 84 **Molecules with ALMA at Planet-forming Scales (MAPS). XIX. Spiral Arms, a Tail, and Diffuse Structures Traced by CO around the GM Aur Disk**
Huang, J., E. A. Bergin, K. I. Öberg, et al.
The Astrophysical Journal Supplement Series, 257, 19
- 83 **Molecules with ALMA at Planet-forming Scales (MAPS). XVIII. Kinematic Substructures in the Disks of HD 163296 and MWC 480**
Teague, R., J. Bae, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 18
- 82 **Molecules with ALMA at Planet-forming Scales (MAPS). XVII. Determining the 2D Thermal Structure of the HD 163296 Disk**
Calahan, J. K., E. A. Bergin, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 17
- 81 **Molecules with ALMA at Planet-forming Scales (MAPS). XVI. Characterizing the Impact of the Molecular Wind on the Evolution of the HD 163296 System**
Booth, A. S., B. Tabone, J. D. Ilee, et al.
The Astrophysical Journal Supplement Series, 257, 16
- 80 **Molecules with ALMA at Planet-forming Scales (MAPS). XV. Tracing Protoplanetary Disk Structure within 20 au**
Bosman, A. D., E. A. Bergin, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 15
- 79 **Molecules with ALMA at Planet-forming Scales (MAPS). XIV. Revealing Disk Substructures in Multiwavelength Continuum Emission**
Sierra, A., L. M. Pérez, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 14
- 78 **Molecules with ALMA at Planet-forming Scales (MAPS). XIII. HCO⁺ and Disk Ionization Structure**
Aikawa, Y., G. Cataldi, Y. Yamato, et al.
The Astrophysical Journal Supplement Series, 257, 13
- 77 **Molecules with ALMA at Planet-forming Scales (MAPS). XII. Inferring the C/O and S/H Ratios in Protoplanetary Disks with Sulfur Molecules**
Le Gal, R., K. I. Öberg, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 12
- 76 **Molecules with ALMA at Planet-forming Scales (MAPS). XI. CN and HCN as Tracers of Photochemistry in Disks**
Bergner, J. B., K. I. Öberg, V. V. Guzmán, et al.
The Astrophysical Journal Supplement Series, 257, 11
- 75 **Molecules with ALMA at Planet-forming Scales (MAPS). X. Studying Deuteration at High Angular Resolution toward Protoplanetary Disks**
Cataldi, G., Y. Yamato, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 10

- 74 **Molecules with ALMA at Planet-forming Scales (MAPS). IX. Distribution and Properties of the Large Organic Molecules HC₃N, CH₃CN, and c-C₃H₂**
Ilee, J. D., C. Walsh, A. S. Booth, et al.
The Astrophysical Journal Supplement Series, 257, 9
- 73 **Molecules with ALMA at Planet-forming Scales (MAPS). VIII. CO Gap in AS 209-Gas Depletion or Chemical Processing?**
Alarcón, F., A. D. Bosman, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 8
- 72 **Molecules with ALMA at Planet-forming Scales (MAPS). VII. Substellar O/H and C/H and Superstellar C/O in Planet-feeding Gas**
Bosman, A. D., F. Alarcón, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 7
- 71 **Molecules with ALMA at Planet-forming Scales (MAPS). VI. Distribution of the Small Organics HCN, C₂H, and H₂CO**
Guzmán, V. V., J. B. Bergner, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 6
- 70 **Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions**
Zhang, K., A. S. Booth, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 5
- 69 **Molecules with ALMA at Planet-forming Scales (MAPS). IV. Emission Surfaces and Vertical Distribution of Molecules**
Law, C. J., **R. Teague**, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 4
- 68 **Molecules with ALMA at Planet-forming Scales (MAPS). III. Characteristics of Radial Chemical Substructures**
Law, C. J., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 3
- 67 **Molecules with ALMA at Planet-forming Scales (MAPS). II. CLEAN Strategies for Synthesizing Images of Molecular Line Emission in Protoplanetary Disks**
Czekala, I., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 2
- 66 **Molecules with ALMA at Planet-forming Scales (MAPS). I. Program Overview and Highlights**
Öberg, K. I., V. V. Guzmán, C. Walsh, et al.
The Astrophysical Journal Supplement Series, 257, 1
- 65 **The First Detection of CH₂CN in a Protoplanetary Disk**
Canta, A., **R. Teague**, R. Le Gal, et al.
The Astrophysical Journal, 922, 62

- 64 **Mapping the 3D Kinematical Structure of the Gas Disk of HD 169142**
Yu, H., **R. Teague**, J. Bae, et al.
The Astrophysical Journal, 920, L33
- 63 **The Chemical Inventory of the Planet-hosting Disk PDS 70**
Facchini, S., **R. Teague**, J. Bae, et al.
The Astronomical Journal, 162, 99
- 62 **A Circumplanetary Disk around PDS70c**
Benisty, M., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 916, L2
- 61 **Limits on Millimeter Continuum Emission from Circumplanetary Material in the DSHARP Disks**
Andrews, S. M., W. Elder, S. Zhang, et al.
The Astrophysical Journal, 916, 51
- 60 **The Architecture of the V892 Tau System: The Binary and Its Circumbinary Disk**
Long, F., S. M. Andrews, J. Vega, et al.
The Astrophysical Journal, 915, 131
- 59 **Investigating the Relative Gas and Small Dust Grain Surface Heights in Protoplanetary Disks**
Rich, E. A., **R. Teague**, J. D. Monnier, et al.
The Astrophysical Journal, 913, 138
- 58 **Vortex-like kinematic signal, spirals, and beam smearing effect in the HD 142527 disk**
Boehler, Y., F. Ménard, C. M. T. Robert, et al.
Astronomy and Astrophysics, 650, A59
- 57 **Observational Signature of Tightly Wound Spirals Driven by Buoyancy Resonances in Protoplanetary Disks**
Bae, J., **R. Teague**, & Z. Zhu
The Astrophysical Journal, 912, 56
- 56 **An Atacama Large Millimeter/submillimeter Array Survey of Chemistry in Disks around M4-M5 Stars**
Pegues, J., K. I. Öberg, J. B. Bergner, et al.
The Astrophysical Journal, 911, 150
- 55 **The TW Hya Rosetta Stone Project IV: A Hydrocarbon-rich Disk Atmosphere**
Cleeves, L. I., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal, 911, 29

- 54 **A highly non-Keplerian protoplanetary disc. Spiral structure in the gas disc of CQ Tau**
Wölfer, L., S. Facchini, N. T. Kurtovic, et al.
Astronomy and Astrophysics, 648, A19
- 53 **ALMA CN Zeeman Observations of AS 209: Limits on Magnetic Field Strength and Magnetically Driven Accretion Rate**
Harrison, R. E., L. W. Looney, I. W. Stephens, et al.
The Astrophysical Journal, 908, 141
- 52 **Dynamical Masses and Stellar Evolutionary Model Predictions of M Stars**
Pegues, J., I. Czekala, S. M. Andrews, et al.
The Astrophysical Journal, 908, 42
- 51 **The TW Hya Rosetta Stone Project. III. Resolving the Gaseous Thermal Profile of the Disk**
Calahan, J. K., E. Bergin, K. Zhang, et al.
The Astrophysical Journal, 908, 8
- 50 **The TW Hya Rosetta Stone Project. II. Spatially Resolved Emission of Formaldehyde Hints at Low-temperature Gas-phase Formation**
Terwisscha van Scheltinga, J., M. R. Hogerheijde, L. I. Cleeves, et al.
The Astrophysical Journal, 906, 111
- 49 **The TW Hya Rosetta Stone Project. I. Radial and Vertical Distributions of DCN and DCO⁺**
Öberg, K. I., L. I. Cleeves, J. B. Bergner, et al.
The Astronomical Journal, 161, 38
- 48 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). V. Sample, overview, and demography of disk molecular emission**
Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 645, A145
- 47 **Chemical Evolution in a Protoplanetary Disk within Planet Carved Gaps and Dust Rings**
Alarcón, F., **R. Teague**, K. Zhang, et al.
The Astrophysical Journal, 905, 68
- 46 **Predicting the Kinematic Evidence of Gravitational Instability**
Hall, C., R. Dong, **R. Teague**, et al.
The Astrophysical Journal, 904, 148
- 45 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). III. The interplay between gas and dust in the protoplanetary disk of DG Tau**
Podio, L., A. Garufi, C. Codella, et al.
Astronomy and Astrophysics, 644, A119

2020

17 PAPERS

- 44 **GoFish: Molecular line detections in protoplanetary disks**
Teague, R.
Astrophysics Source Code Library, ascl:2011.016
- 43 **ALMA and VLA Observations of EX Lupi in Its Quiescent State**
White, J. A., Á. Kóspál, A. G. Hughes, et al.
The Astrophysical Journal, 904, 37
- 42 **Visualizing the Kinematics of Planet Formation**
Collaboration, D. D., P. J. Armitage, J. Bae, et al.
arXiv e-prints, arXiv:2009.04345
- 41 **Low-level Carbon Monoxide Line Polarization in Two Protoplanetary Disks: HD 142527 and IM Lup**
Stephens, I. W., M. Fernández-López, Z. Li, et al.
The Astrophysical Journal, 901, 71
- 40 **The Excitation Conditions of CN in TW Hya**
Teague, R. & R. Loomis
The Astrophysical Journal, 899, 157
- 39 **Annular substructures in the transition disks around LkCa 15 and J1610**
Facchini, S., M. Benisty, J. Bae, et al.
Astronomy and Astrophysics, 639, A121
- 38 **A three-dimensional view of Gomez's hamburger**
Teague, R., M. R. Jankovic, T. J. Haworth, et al.
Monthly Notices of the Royal Astronomical Society, 495, 451
- 37 **The efficiency of dust trapping in ringed protoplanetary discs**
Rosotti, G. P., **R. Teague**, C. Dullemond, et al.
Monthly Notices of the Royal Astronomical Society, 495, 173
- 36 **Hints of a Population of Solar System Analog Planets from ALMA**
Long, D. E., K. Zhang, **R. Teague**, et al.
The Astrophysical Journal, 895, L46
- 35 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). I. CO, CS, CN, and H₂CO around DG Tau B**
Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 636, A65
- 34 **A Multifrequency ALMA Characterization of Substructures in the GM Aur Protoplanetary Disk**
Huang, J., S. M. Andrews, C. P. Dullemond, et al.
The Astrophysical Journal, 891, 48

2019

12 PAPERS

- 33 **Spiral arms in the protoplanetary disc HD100453 detected with ALMA: evidence for binary-disc interaction and a vertical temperature gradient**
Rosotti, G. P., M. Benisty, A. Juhász, et al.
Monthly Notices of the Royal Astronomical Society, 491, 1335
- 32 **Accretion disks around young stars: the cradles of planet formation**
Semenov, D. A. & **R. D. Teague**
Europhysics News, 51, 29
- 31 **Tracing the physical conditions of planet formation with molecular excitation**
Teague, R.
Laboratory Astrophysics: From Observations to Interpretation, 350, 181
- 30 **Meridional flows in the disk around a young star**
Teague, R., J. Bae, & E. A. Bergin
Nature, 574, 378
- 29 **Spiral Structure in the Gas Disk of TW Hya**
Teague, R., J. Bae, J. Huang, et al.
The Astrophysical Journal, 884, L56
- 28 **An Ideal Testbed for Planet-Disk Interaction: Two Giant Protoplanets in Resonance Shaping the PDS 70 Protoplanetary Disk**
Bae, J., Z. Zhu, C. Baruteau, et al.
The Astrophysical Journal, 884, L41
- 27 **ALMA observations of A0620-00: fresh clues on the nature of quiescent black hole X-ray binary jets**
Gallo, E., **R. Teague**, R. M. Plotkin, et al.
Monthly Notices of the Royal Astronomical Society, 488, 191
- 26 **GoFish: Fishing for Line Observations in Protoplanetary Disks**
Teague, R.
The Journal of Open Source Software, 4, 1632
- 25 **Detection of Continuum Submillimeter Emission Associated with Candidate Protoplanets**
Isella, A., M. Benisty, **R. Teague**, et al.
The Astrophysical Journal, 879, L25
- 24 **Statistical Uncertainties in Moment Maps of Line Emission**
Teague, R.
Research Notes of the American Astronomical Society, 3, 74
- 23 **Planet formation — The case for large efforts on the computational side**
Lyra, W., T. Haworth, B. Bitsch, et al.
Bulletin of the American Astronomical Society, 51, 129

2018

7 PAPERS

- 22 **Realizing the Unique Potential of ALMA to Probe the Gas Reservoir of Planet Formation**
Cleeves, I., R. Loomis, **R. Teague**, et al.
Bulletin of the American Astronomical Society, 51, 81
- 21 **Line Ratios Reveal N_2H^+ Emission Originates above the Midplane in TW Hydrae**
Schwarz, K. R., **R. Teague**, & E. A. Bergin
The Astrophysical Journal, 876, L13
- 20 **Highly structured disk around the planet host PDS 70 revealed by high-angular resolution observations with ALMA**
Keppler, M., **R. Teague**, J. Bae, et al.
Astronomy and Astrophysics, 625, A118
- 19 **eddy: Extracting Protoplanetary Disk Dynamics with Python**
Teague, R.
The Journal of Open Source Software, 4, 1220
- 18 **Evidence for a Vertical Dependence on the Pressure Structure in AS 209**
Teague, R., J. Bae, T. Birnstiel, et al.
The Astrophysical Journal, 868, 113
- 17 **A Robust Method to Measure Centroids of Spectral Lines**
Teague, R. & D. Foreman-Mackey
Research Notes of the American Astronomical Society, 2, 173
- 16 **Temperature, Mass, and Turbulence: A Spatially Resolved Multiband Non-LTE Analysis of CS in TW Hya**
Teague, R., T. Henning, S. Guilloteau, et al.
The Astrophysical Journal, 864, 133
- 15 **Chemistry in disks. XI. Sulfur-bearing species as tracers of protoplanetary disk physics and chemistry: the DM Tau case**
Semenov, D., C. Favre, D. Fedele, et al.
Astronomy and Astrophysics, 617, A28
- 14 **A Kinematical Detection of Two Embedded Jupiter-mass Planets in HD 163296**
Teague, R., J. Bae, E. A. Bergin, et al.
The Astrophysical Journal, 860, L12
- 13 **Turbulence in the TW Hya Disk**
Flaherty, K. M., A. M. Hughes, **R. Teague**, et al.
The Astrophysical Journal, 856, 117

2017

7 PAPERS

- 12 **ALMA continuum observations of the protoplanetary disk AS 209. Evidence of multiple gaps opened by a single planet**
Fedele, D., M. Tazzari, R. Booth, et al.
Astronomy and Astrophysics, 610, A24
- 11 **Radiation Hydrodynamical Turbulence in Protoplanetary Disks: Numerical Models and Observational Constraints**
Flock, M., R. P. Nelson, N. J. Turner, et al.
The Astrophysical Journal, 850, 131
- 10 **The Flying Saucer: Tomography of the thermal and density gas structure of an edge-on protoplanetary disk**
Dutrey, A., S. Guilloteau, V. Piétu, et al.
Astronomy and Astrophysics, 607, A130
- 9 **Multiplicity and disks within the high-mass core NGC 7538IRS1. Resolving cm line and continuum emission at $0.06'' \times 0.05''$ resolution**
Beuther, H., H. Linz, T. Henning, et al.
Astronomy and Astrophysics, 605, A61
- 8 **On the methanol emission detection in the TW Hya disc: the role of grain surface chemistry and non-LTE excitation**
Parfenov, S. Y., D. A. Semenov, T. Henning, et al.
Monthly Notices of the Royal Astronomical Society, 468, 2024
- 7 **Three Radial Gaps in the Disk of TW Hydrae Imaged with SPHERE**
van Boekel, R., T. Henning, J. Menu, et al.
The Astrophysical Journal, 837, 132
- 6 **A Surface Density Perturbation in the TW Hydrae Disk at 95 au Traced by Molecular Emission**
Teague, R., D. Semenov, U. Gorti, et al.
The Astrophysical Journal, 835, 228
- 5 **Tracing the earliest stages of planet formation through modelling and sub-mm observations**
Teague, R.
Ph.D. Thesis

2016

3 PAPERS

- 4 **Grand Challenges in Protoplanetary Disc Modelling**
Haworth, T. J., J. D. Ilee, D. H. Forgan, et al.
Publications of the Astronomical Society of Australia, 33, e053
- 3 **Inferring the evolutionary stages of the internal structures of NGC 7538 S and IRS1 from chemistry**
Feng, S., H. Beuther, D. Semenov, et al.
Astronomy and Astrophysics, 593, A46

2015

1 PAPER

- 2 **Measuring turbulence in TW Hydrae with ALMA: methods and limitations**
Teague, R., S. Guilloteau, D. Semenov, et al.
Astronomy and Astrophysics, 592, A49
- 1 **Chemistry in disks. IX. Observations and modelling of HCO^+ and DCO^+ in DM Tauri**
Teague, R., D. Semenov, S. Guilloteau, et al.
Astronomy and Astrophysics, 574, A137